Lecture 5: Addendum

Functional Analysis Survival Guide

Biostar Quote Of The Day Why does each GO enrichment method give different results?

> *I'm new to GO terms. In the beginning it was fun, as long as I stuck to one algorithm.*
>
> *But then I found that there are many out there, each with its own advantages and caveats (the quality of graphic representation, for instance)*
>
> *As a biologist, what should I trust? Deciding on this or that algorithm may change the whole story!*

GO Enrichment Caveats

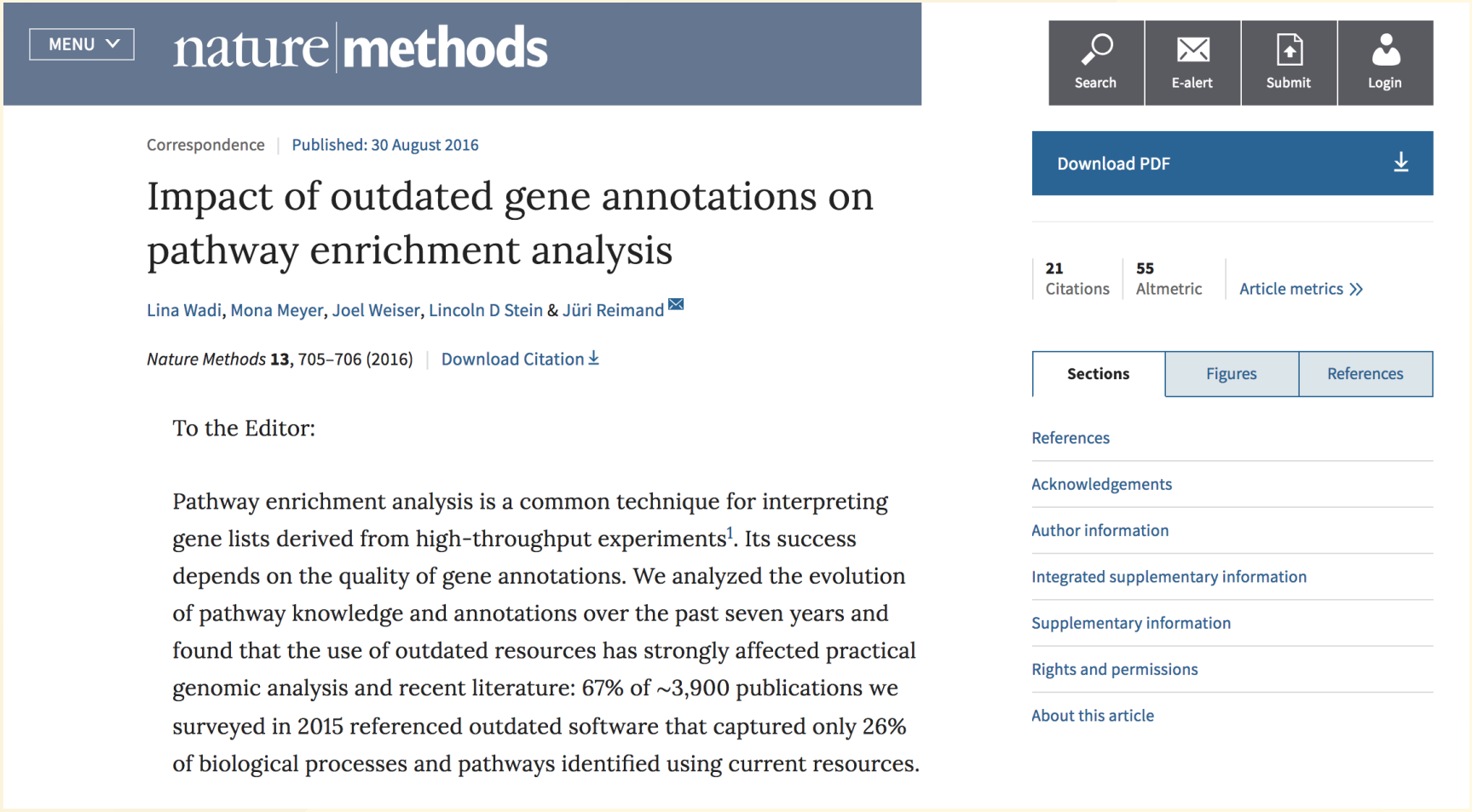
The data in incomplete!

Every month genes get annotated and perhaps re-annotated

How incomplete? -> Surprisingly hard to find out the rate of growth. *This is an interesting bioinformatics project on its own.*

How have scientific dicoveries change as more information becomes available?

Impact of outdated annotations



Read this paper for more background on enrichment.

You have to go full X-Files Mode

Step 1



Step 2



Gene Ontology Statistics is Flakey

Annotations are not distributed randomly

(or by a "representative" selection)

Annotations are not discovered randomly

(or by a "representative" selection)

Gene Enrichment Studies are a great First Step

But they are just a First Step

Your favorite tool might not work tomorrow!

The flaws of scientific funding are clearly exposed.

There is no support for *keeping* a tool working.

Scientists do not like to fund projects that *merely* support existing tools.

Hence just about all tools wither within years.

How to understand a Gene List

1. Understand the known functions
2. Use different tools, understand what each does
3. Explore the "neighbors" of the terms
4. Find information not in Gene Ontology
5. Don't be afraid to connect the two
6. Rebel against the tyranny of the P-Values.